Proc. of the International Conference on Electrical, Computer and Energy Technologies (ICECET 2022)
20-22 July 2022, Prague-Czech Republic

A Novel Gain Enhanced Folded Cascode OPAMP in 28nm CMOS Technology

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2022 International Conference on Electrical, Computer and Energy Technologies (ICECET) | 978-1-6654-7087-2/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICECET55527.2022.9872766

***Abstract*— Folded cascade Opamp is the preferred first stage choice in a typical two-stage opamp due to relaxed headroom and better output swing. Traditional Folded cascade opamp gain and noise are compromised due to the folding transistors. In this work, FVF assisted differential pair has been proposed which extracts the signal and injects it onto the gate of the folding devices, hence they will contribute gain and noise will be reduced. A prototype was developed in 28nm TSMC CMOS technology and post-layout simulation results were performed. The proposed opamp achieves 68dB DC gain, which is 18dB higher than the conventional one. Also, the proposal exhibits 20% improvement in the input-referred noise for the same power consumption. Further, the circuit consumes 33µW of power from a 1V power supply and occupies a 78*46µm2 silicon area.**

Keywords—Noise, gain, folded cascode, mismatch, distortion, phase margin.

I. INTRODUCTION

Internet of things (IOT) and smart sensors are getting a lot of traction in the present era. The main bottleneck of these circuit realizations is to have an accurate reference voltage/current generator, which is through a Bandgap reference (BGR). The accuracy requirement of the BGR is becoming tight due to the high precision ADC development in sub-micron CMOS technologies, mainly under the low supply voltage domain [1]. Unfortunately, the threshold voltage (v_{th}) is not scaling at the same pace as vdd increment hence opamps are suffering from poor headrooms. So designing high gain Opamps became very challenging.

Recently Folded cascade (FC) opamp has gotten a lot of traction compared to the telescopic cascade because of its relatively larger signal swing and gain compared to the telescopic opamp [2]. Also PMOS input pair opamp is the major choice due to its better frequency response and lower flicker noise. Unfortunately, FC opamp voltage gain in 45nm and lower technology nodes is getting lower than 40dB due to the lower o/p impedance (g_{ds}). There has been a lot of research done to develop high-performance FC opamps. [3] proposed a multi current mirror OTA to enhance the gain and slew rate. [4] proposes a recycled architecture, which enhances the gain and Unity Gain Bandwidth (UGB) by splitting the current in the differential pair. [4] is the modified version of [5], to embedded the class-AB output stage. Also [6] describes the double recycling technique to enhance gain further.

The rest of the paper has been organized as follows. Section II analyses the conventional folded cascade and its disadvantages. Section III explains the proposed FC opamp qualitatively and quantitatively. Section IV summarizes the post-layout simulation results and state of art comparison.

II. CONVENTIONAL FOLDED CASCODE OPAMP

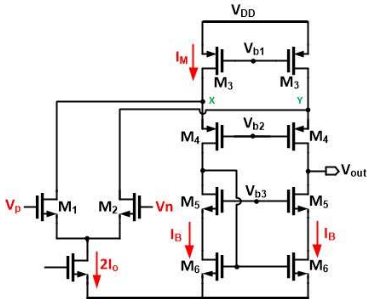

Fig. 1. Conventional NMOS input folded Cascode Opamp.

Fig:1 depicts the conventional NMOS input FC opamp. M_1,M_2 form the differential pair and M_4,M_5,M_6 form the cascade load and output current mirror. FC opamp can be viewed as NMOS differential amplifier signal current injected into the pmos current buffer (common gate amp). As shown in fig:1, the differential pair signal will be injected into M_4 source, hence both M_1,M_4 bias currents have to be balanced by M_3 so it will carry a much higher current than other transistors. V_{b1},V_{b2},V_{b3} cascode bias voltages will be generated from a typical high swing cascade network (not shown here for simplicity) [3]. The output swing is not directly related to the input common-mode voltage because of the folding nature at the node x,y and hence output stage swing will be high. The differential output swing can be expressed as $2*(V_{dd}-4V_{ov})$, which is relatively high compared to the telescopic version. Also designing unity gain voltage buffers is relatively easy because of the above reason. Though higher swing is the main advantage, this FC opamp has several disadvantages as listed below [4]. It has poor low-frequency voltage gain because the output impedance is lower due to the g_{ds1} loading at nodes x,y. The dc voltage can be expressed as follows. Where gm, gds represent the small-signal transconductance and output conductance respectively.

$$A_V = \frac{g_{m1}}{\frac{g_{ds5}g_{ds6}}{g_{m5}} + \frac{(g_{ds3}+g_{ds1})g_{ds4}}{g_{m4}}} \qquad (1)$$

The second disadvantage is the higher power consumption. From the fig:1, it is clear that M_3 should carry sum of the M_1 and M_4 current which means $I_M=I_B+I_0$. The +ve and –ve slew rate of the FC opamp can be expressed as (2), hence to achieve symmetric slewrate $I_B>=2I_0$, otherwise output signal will be distorted heavily which is highly undesirable in switched capacitor amplifier.

$$SR_+ = \frac{2I_0}{C_L} \qquad (2)$$

Overall, the power consumption is much higher than the telescopic opamp (mainly due to the folding nature). The input-referred power spectral density (PSD) can be expressed as shown in equation (3).

$$V_n^2 = \frac{4KT\gamma}{g_{m1}}\left(1 + \frac{g_{m3}}{g_{m1}} + \frac{g_{m6}}{g_{m1}}\right) \qquad (3)$$

The majority of the thermal noise is contributed by the M_1,M_3,M_6 and cascade transistor noise is suppressed by the intrinsic gain of the transistor hence they were not included in (3). In the conventional telescopic opamp there is no M_6 contribution in the noise. Since M_3 carries a much higher current than any other transistor, g_{m3} is high and hence it results in more noise. This is considered the major disadvantage of the FC opamp compared to the telescopic, especially in high-resolution applications like continuous-time sigma-delta ADC/DAC applications [6].

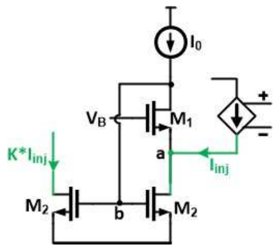

Fig. 2. Flipped Voltage Follower (FVF) assisted differential pair

In a summary Folded cascade opamp is interesting architecture from the signal swing point of view but it has poor dc gain and higher input-referred noise while consuming more power [9].

III. Proposed Technique

This section explains how to improve the DC gain with improved input-referred noise. The motivation for the present proposal is as follows. The folding transistor (M_3 in fig:1) carries more current without contributing any signal current (which means no role in voltage gain) while contributing more noise. One way of utilizing g_{m3} is to somehow inject the input differential voltage signal onto the gates of the M_3, which requires two steps. The first step is to separate some of the input signal current from the differential pair and the second step is to inject on top of the gate of M_3. Fig:2 depicts the idea of the first step, which is Flipped Voltage Follower (FVF) assisted differential pair [7][8]. It is a voltage buffer formed with M_1,M_2. The shunt-shunt feedback from the M_1 drain to the M_2 gate reduces the o/p impedance from $\frac{1}{g_{m1}}$ to $\frac{g_{ds2}}{g_{m1}g_{m2}}$. If there is any signal injected into node A shown in fig:3, it has to flow through M_2 only because M_1 is biased with the fixed current source I_B. Hence by using a current mirror formed M_2, one can extract/amplify the signal current. In the fig:2 VCCS represents the differential pair (signal current injection). The impedance at node a is very low such that it can act as a tail node in the differential pair [10][11]. Based on this concept, the proposed opamp will be synthesized as follows.

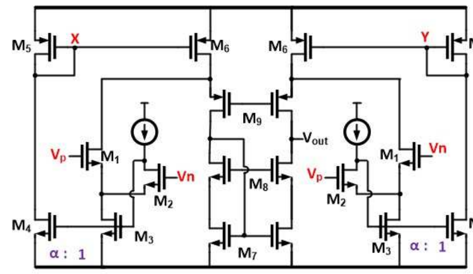

Fig. 3. Proposed Gain enhanced Folded Cascode opamp

Fig:3 shows the proposed gain enhanced FC opamp. Here M_1,M_2,M_3 form FVF assisted differential pair. Compared to the Fig:2, the only difference in the FVF circuit is M_1,M_2 both which have been exited by the differential signals V_P,V_N. M_1,M_3 carry small signal currents proportional to $g_{m1}(V_p-V_n)$. The signal current of M_1 will be injected into the M_9 source exactly like the traditional FC cascade opamp. The main difference in the proposal is Folding transistor M_6 gate will be exited by the signal also such that it contributes signal current instead of only supporting the bias current. M_4, M_5 transistors form an amplifier which converts M_3 current into the voltage at node X, the current mirror gain ratio is α. The voltage at node X and current through M_6 can be expressed as follows.

$$V_X = \alpha \frac{g_{m1}}{g_{m5}}(V_P - V_N) \qquad (4)$$

$$I_{M6} = g_{m6}\alpha \frac{g_{m1}}{g_{m5}}(V_P - V_N) \qquad (5)$$

The voltage gain of the proposed amplifier can be expressed as follows. It reveals that the proposed amplifier gain has been improved by $1 + \alpha\frac{g_{m6}}{g_{m5}}$ times compared to the traditional FC amplifier, which is a substantial improvement, and gain boosting can be adjusted by α. Hence α will be considered as the main design parameter for this proposal.

$$A_V = \frac{g_{m1}\left(1+\alpha\frac{g_{m6}}{g_{m5}}\right)}{\frac{g_{ds5}g_{ds6}}{g_{m5}}+\frac{(g_{ds3}+g_{ds1})g_{ds4}}{g_{m4}}} \qquad (6)$$

Due to the increase in the effective transconductance, the input-referred noise will also decrease proportionally. Equation (7) shows the PSD of the input-referred noise, in comparison to the conventional FC (eq (3)), this is a significant reduction. Intuitively we explain this reduction as more number of signal generating transistors while keeping the number of noise generating devices more.

$$V_n^2 = \frac{4KT\gamma}{g_{m1}\left(1+\alpha\frac{g_{m6}}{g_{m5}}\right)}\left(1+\frac{g_{m3}}{g_{m1}}+\frac{g_{m6}}{g_{m1}}\right) \qquad (7)$$

α can't be increased very high due to stability The answer is no, because it will affect stability[12]. Looking at the fig:3, the parasitic capacitance at node x will create an additional pole in the transfer function and compromises the stability. Hence increasing α will improve the dc gain while decreasing the Phase margin (PM). Equation (8) shows the closed-form solution for PM as a function of α, where $W_u$represents the Unity Gain Bandwidth (UGB) and C_X is the capacitance at node-x.

$$PM = 90^0 - \tan^{-1}\left(\frac{W_u}{\frac{g_{m5}}{C_X}}\right) \qquad (8)$$

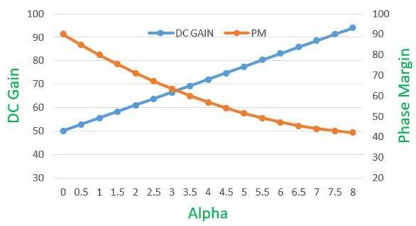

Fig. 4. DC gain and PM versus α

Fig:4 shows the DC gain and PM versus α of the model opamp. As explained in the theory gain increases with α, starting from 50-95dB while varying α from 0 to 8, whereas the PM decreases from 90-40. The optimal value α of lies arod 3 in this particular implementation, beyond this value, step response will have significant rining.

IV. SIMULATION RESULTS.

The proposed technique has been implemented in 1poly 10 metal 28nm CMOS technology. Conventional and the proposed FC opamp have been designed for proper comparison. Designed with g_m/i_d ratio of 15 for the differential pair and 8 for the current mirrors [13]. The opamp was designed with 1V power supply and draws 33uA in a Fast process corner and 125^0C temperature. Fig:5 shows the frequency response of both opamps in unity gain configuration, conventional opamp has 50dB gain whereas the proposed has been displayed 68dB (18dB gain boosting), while keeping the UGB around 10MHz. The proposed opamp has a non-dominate pole at 20MHz, so the PM has become 63^0, whereas the conventional opamp has close to 90^0.

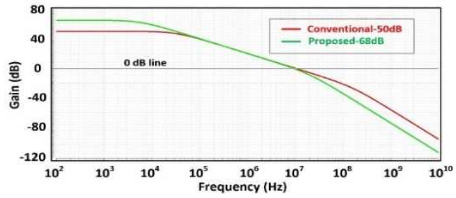

Fig. 5. Frequency response of the Proposed and Conventional Opamp

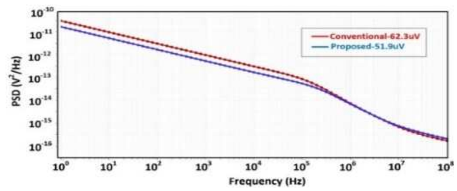

Fig. 6. Simulated noise of the Proposed and Conventional opamp

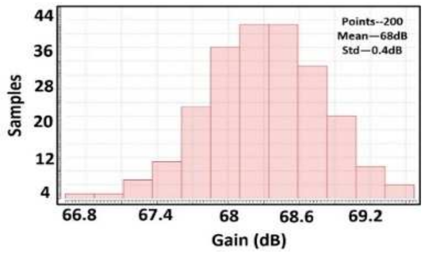

Fig. 7. 200 point Histogram of the DC gain.

Fig:6 shows the simulated input-referred noise of both opamps. As explained in eq (7), the proposed solution has a substantial reduction in the noise due to the contribution of the folding devices g_m. At low frequency, the PSD of the proposed opamp is improved by 4.5dB. The integrated RMS noise of the proposed opamp is 51.9uV, whereas the conventional one resulted in 62.3uV, which is a ~20% improvement. Usually, noise improvement results in SNR enhancement without increasing the power consumption of the opamp. For the first order, the proposed technique gain enhancement is insensitive to the mismatch unlike +ve feedback-based gain techniques like [14]. Fig:7 depicts the 200-point histogram of the DC gain, which shows a 68dB mean and standard deviation of 0.4dB (~0.5%). Fig:8 shows the layout of the proposed circuit, which occupies a $78*46um^2$ active silicon area. Adopted all mismatch reduction techniques like common centroid and interdigitation techniques for layout [15].

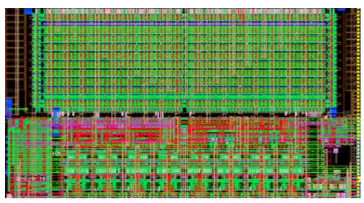

Fig. 8. 200 point Histogram of the DC gain.

Table-I:-Comparision with the state of art

Parameter	Unit	[3]	[4]	[5]	This work
Technology	nm	90	180	65	28
Load Capacitance	pF	5	5.6	5	10
DC gain	dB	59	53.6	62	68
FOM	MHz.pF/µW	204	52	187	498.5

IV. CONCLUSION

In this work, a gain enhanced Folded Cascode opamp has been proposed, which displayed 18dB better DC gain and 20% better input-referred noise.

REFERENCES

[1] A. Awny *et al.*, "A Linear Differential Transimpedance Amplifier for 100-Gb/s Integrated Coherent Optical Fiber Receivers," in *IEEE Transactions on Microwave Theory and Techniques*, vol. 66, no. 2, pp. 973-986, Feb. 2018

[2] R. Nagulapalli, K. Hayatleh, S. Barker, S. Zourob, N. Yassine and S. Sridevi, "A bio-medical compatible self bias opamp in 45nm CMOS technology," *2017 International conference on Microelectronic Devices, Circuits and Systems (ICMDCS)*, 2017, pp. 1-4,

[3] R. Nagulapalli and R. K. Palani, "A Novel 22.7 ppm/0C Voltage mode Sub-Bandgap Reference with robust startup nature," *2021 IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2021, pp. 844-847

[4] R. S. Assaad and J. Silva-Martinez, “The recycling folded cascode: A general enhancement of the folded cascode amplifier,” IEEE J. SolidState Circuits, vol. 44, no. 9, pp. 2535–2542, Sep. 2009

[5] Nagulapalli, R., Hayatleh, K., Barker, S. et al. An OTA gain enhancement technique for low power biomedical applications. Analog Integr Circ Sig Process 95, 387–394 (2018).

[6] R. Nagulapalli, R. K. Palani, K. Hayatleh, N. Yassine and S. Barker, "A 0.82 V Supply and 23.4 ppm/°C Current Mirror Assisted Bandgap Reference," *2021 32nd Irish Signals and Systems Conference (ISSC)*, 2021, pp. 1-4

[7] R. Nagulapalli, K. Hayatleh, S. Barker, S. Zourob, N. Yassine , "A 31 ppm/oC Pure CMOS Bandgap Reference by Exploiting Beta-Multiplier", *VLSI Design and Test. VDAT 2018. Communications in Computer and Information Science*, vol. 892, 2019.

[8] R. Nagulapalli, R. K. Palani, S. Agarwal, S. C. K. Hayatleh and S. Barker, "A 15uW, 12 ppm/°C Curvature Compensated Bandgap in 0.85V Supply," *2021 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2021, pp. 1-4

[9] Mili Anand, Anushree, Jasdeep Dhanoa, "Gain and Gain-bandwidth Enhancement of Recyclic Folded Cascode OTA using Floating Voltage Source", 2022 IEEE Delhi Section Conference (DELCON), pp.1-5, 2022

[10] R. Nagulapalli, R. K. Palani and S. Bhagavatula, "A 24.4 ppm/oC Voltage Mode Bandgap Reference with a 1.05V supply," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, doi: 10.1109/TCSII.2020.3034256.

[11] R. Nagulapalli, K. Hayatleh, S. Barker and B. N. K. Reddy, "A two-stage opamp frequency Compensation technique by splitting the 2nd stage," *2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2020, pp. 1-5

[12] K. Desai, R. Nagulapalli, V. Krishna, R. Palwai, P. K. Venkatesan and V. Khawshe, "High Speed Clock and Data Recovery Circuit with Novel Jitter Reduction Technique," *2010 23rd International Conference on VLSI Design*, 2010, pp. 300-305

[13] M. Kroh *et al.*, "Monolithic Photonic-Electronic Linear Direct Detection Receiver for 56Gbps OOK," *ECOC 2016; 42nd European Conference on Optical Communication*, 2016, pp. 1-3

[14] R. Nagulapalli, S. Zourob, K. Hayatleh, N. Yassine, S. Barker and A. Venkatareddy, "A compact high gain opamp for Bio-medical applications in 45nm CMOStechnology," *2017 2nd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, 2017, pp. 231-235

[15] T. Varun, R. Nagulapalli and I. Raja, "A 82µW Mixed-Mode sub-1V Bandgap reference with 25 ppm/°C Temperature Co-efficient with Simultaneous PTAT Generation," *2021 25th International Symposium on VLSI Design and Test (VDAT)*, 2021, pp. 1-4

[16] A. Awny *et al.*, "23.5 A dual 64Gbaud 10kΩ 5% THD linear differential transimpedance amplifier with automatic gain control in 0.13µm BiCMOS technology for optical fiber coherent receivers," *2016 IEEE International Solid-State Circuits Conference (ISSCC)*, 2016, pp. 406-40

[17] R. Nagulapalli, K. Hayatleh, S. Barker and B. N. K. Reddy, "A Single BJT 10.2 ppm/°C Bandgap Reference in 45nm CMOS Technology," *2020 11th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2020, pp. 1-4

[18] R. Nagulapalli, K. Hayatleh, S. Barker, B. N. K. Reddy and B. Seetharamulu, "A High Frequency CMRR improvement technique for Differential Amplifiers in 45nm CMOS," *2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2019, pp. 1-5

[19] R. Nagulapalli, K. Hayatleh, S. Barker, S. Zourob and A. Venkatreddy, "A Low Voltage Opamp Based on Current Mirror Technique for Bio-medical applications," *2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, 2018, pp. 901-904

[20] A. Awny *et al.*, "A 40 Gb/s Monolithically Integrated Linear Photonic Receiver in a 0.25 µm BiCMOS SiGe:C Technology," in *IEEE Microwave and Wireless Components Letters*, vol. 25, no. 7, pp. 469-471, July 2015

[21] R. Nagulapalli, K. Hayatleh, S. Barker, S. Zourob, N. Yassine and B. N. K. Reddy, "High Performance Circuit Techniques for Nueral Front-End Design in 65nm CMOS," *2018 9th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, 2018, pp. 1-4

[22] R. Nagulapalli, S. Raparthy and A. Venkatareddy, "A low offset instrumentation amplifier for biomedical applications," *2017 International Conference on Circuits, Controls, and Communications (CCUBE)*, 2017, pp. 191-194

[23] R. Nagulapalli, K. Hayatleh, S. Barker, A. A. Tammam, P. Georgiou and F. J. Lidgey, A 0.55 v bandgap reference with a 59 ppm/◦c temperature coefficient, J. Circuits Syst. Comput. 28 (2019)

[24] R. Nagulapalli, K. Hayatleh, S. Barker, P. Georgiou and F. J. Lidgey , A high value, linear and tunable CMOS pseudo resistor for bio-medical applications, J. Circuits, Syst. Comput. 28 (2018)

[25] R. Nagulapalli, K. Hayatleh, S. Barker, B. Yassine, S. Zourob, S. Raparthy and N. Yassine, A start-up assisted fully differential folded cascode opamp, J. Circuits Syst. Comput. 28 (2019)

[26] P. Mishra *et al.*, "8.7 A 112Gb/s ADC-DSP-Based PAM-4 Transceiver for Long-Reach Applications with >40dB Channel Loss in 7nm FinFET," *2021 IEEE International Solid- State Circuits Conference (ISSCC)*, 2021, pp. 138-140

[27] R. Nagulapalli, K. Hayatleh, S. Barker, A. A. Tammam, N. Yassine, B. Yassine and M. Ben Esmael , A low noise amplifier suitable for biomedical recording analog front-end in 65nm CMOS technology, *J. Circuits Syst. Comput.* **28** (2019)

[28] R. Nagulapalli, K. Hayatleh, S. Barker, S. Zourob, N. Yassine and S. Sridevi, "A Microwatt Low Voltage Bandgap Reference for Bio-medical Applications," *2017 International Conference on Recent Advances in Electronics and Communication Technology (ICRAECT)*, 2017, pp. 61-65